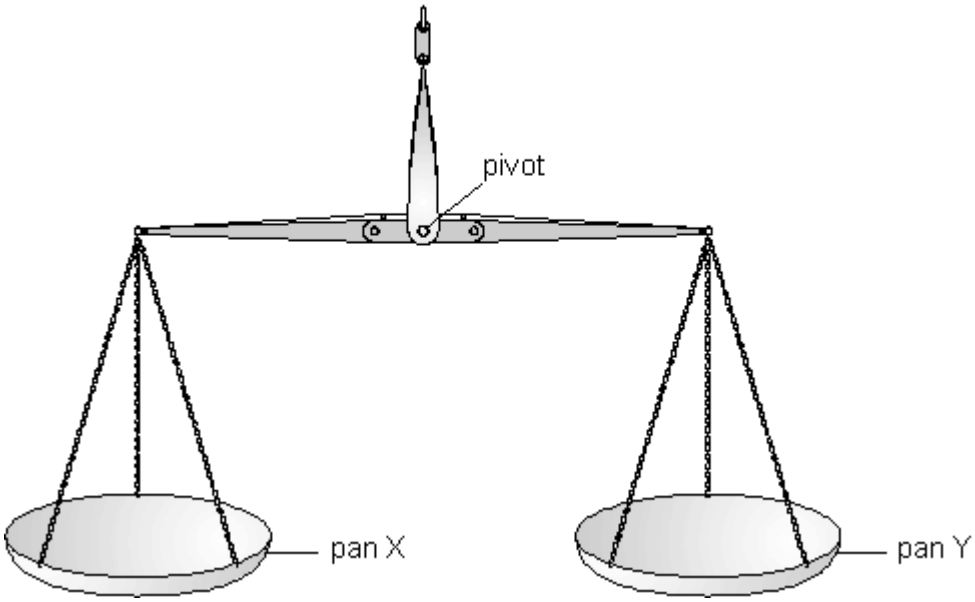
Q1. The drawing shows a very old set of scales. It can be used to check the weights of silver coins.



(a) Rema puts a silver coin in pan X. There is nothing in pan Y.
In which direction does pan X move?

……

1 mark

(b) The table shows the weights of five silver coins.

Silver coin	weight in mN
A	106
B	112
C	98
D	112
E	120

Rema puts one coin in each pan of the scales. Which **two** coins will make the scales balance? Give the letters.

…… and ……

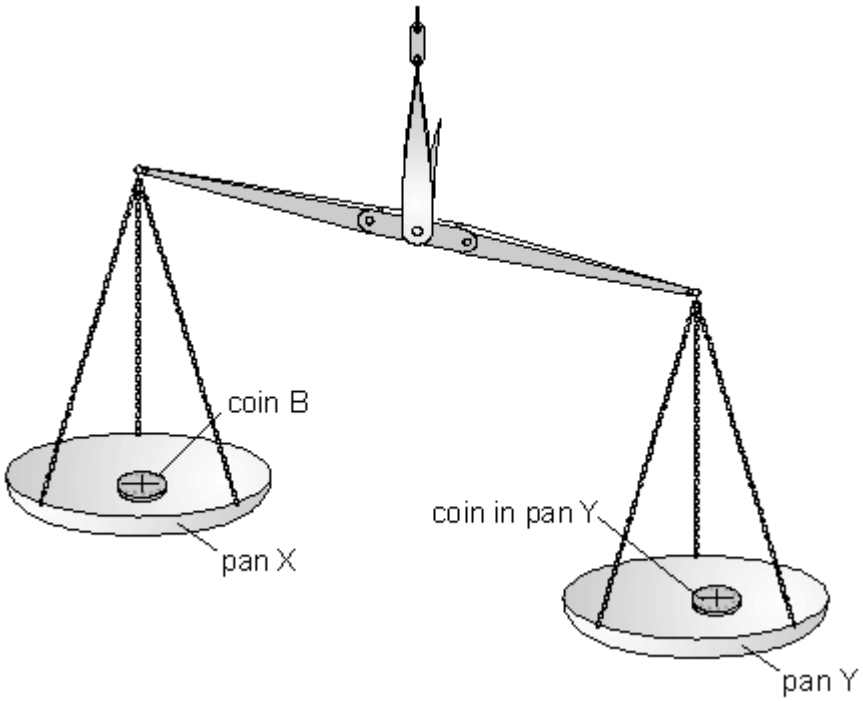
1 mark

(c) Coin A is placed in pan X, and coin C is placed in pan Y.
In which direction does pan X move?

……...

1 mark

(d) In another experiment, coin B is placed in pan X, and one of the other coins is placed in pan Y. Pan X goes **up**.



Which coin is in pan Y? Give the letter.

…….

1 mark

(e) Rema knows the weights of the five silver coins in the table. She also has a gold coin. Explain how she could use the coins and the scales to find the approximate weight of the gold coin.

…….

…….

…….

…….

2 marks
Maximum 6 marks

Q1.

(a) down

1 (L3)

(b) B and D

letters may be in either order
***both** letters are required for the mark*
accept 112 and 112

1 (L3)

(c) down

1 (L3)

(d) E

1 (L3)

(e) any **two** from

- put the gold coin in one pan and a silver coin **or** coins in the other
- add **or** replace **or** remove silver coins
- see which pan goes up **or** see which way the scales move
  accept 'see if it balances'
- it is heavier than the silver coins which go up
- it is lighter than the silver coins which go down
  accept 'try each silver coin to see which one balances the gold coin' for two marks

2 (L4)

[6]